MATHEMATICS & SIGNALS

Complex Number Conversions:

- Rectangular to Polar: $r = \sqrt{x^2 + y^2}$, $\theta = \tan^{-1}\left(\frac{y}{x}\right)$
- Polar to Rectangular: $x = r\cos(\theta)$, $y = r\sin(\theta)$

System Impedances ($s = j\omega$):

- Resistor: $Z_R = R$
- Capacitor: $Z_C = \frac{1}{sC} = \frac{1}{j\omega C} = -j\frac{1}{\omega C}$
- Inductor: $Z_L = sL = j\omega L$

RMS Values (Standard Sinusoids):

- $V_{RMS} = \frac{V_{peak}}{\sqrt{2}}$ • $I_{RMS} = \frac{I_{peak}}{\sqrt{2}}$

CALCULUS RULES

Derivatives:

$\frac{d}{dt}(t^n) = nt^{n-1}$

$\frac{d}{dt}(e^{at}) = ae^{at}$

$\frac{d}{dt}(\sin\omega t) = \omega\cos\omega t$

$\frac{d}{dt}(\cos\omega t) = -\omega\sin\omega t$

Integrals (omit $+C$):

$\int t^n dt = \frac{t^{n+1}}{n+1}$

$\int e^{at} dt = \frac{1}{a}e^{at}$

$\int \sin\omega t dt = -\frac{1}{\omega}\cos\omega t$

$\int \cos\omega t dt = \frac{1}{\omega}\sin\omega t$

PARTIAL FRACTION DECOMPOSITION

Used to invert complex $F(s)$ into time-domain components.

- **Distinct Real Roots:** $\frac{K}{(s-a)(s-b)} = \frac{A}{s-a} + \frac{B}{s-b}$

Heaviside Cover-up: $A = (s-a)F(s)|_{s=a}$

- **Repeated Roots:** $\frac{K}{(s-a)^2} = \frac{A}{(s-a)^2} + \frac{B}{s-a}$
- **Complex Roots:** Complete the square for the denominator: $s^2 + cs + d \implies (s-a)^2 + b^2$.

Expand as: $\frac{A(s-a)}{(s-a)^2+b^2} + \frac{B(b)}{(s-a)^2+b^2}$.

Inverse transforms directly to: $Ae^{at}\cos(bt) + Be^{at}\sin(bt)$.

CIRCUIT ANALYSIS & OP-AMPS

Transfer Functions ($H(s)$):

The ratio of the Laplace transform of the output to the input, assuming zero initial conditions: $H(s) = \frac{V_{out}(s)}{V_{in}(s)}$.

- **Poles & Zeros:** Roots of the numerator are **zeros** (frequencies where output is 0). Roots of the denominator are **poles** (these dictate system stability and the natural, transient response).
- **Frequency Response:** Substitute $s = j\omega$ into $H(s)$ to evaluate the system's magnitude $|H(j\omega)|$ (Gain) and phase shift $\angle H(j\omega)$ at specific frequencies.
- *Tip:* Treat capacitors and inductors as complex impedances (Z) and use standard DC analysis techniques (voltage dividers, node voltage) to solve for V_{out} in terms of V_{in}.

Standard Op-Amp Configurations (Ideal):

- Non-Inverting: $\frac{V_{out}}{V_{in}} = 1 + \frac{R_f}{R_{in}}$
- Inverting: $\frac{V_{out}}{V_{in}} = -\frac{R_f}{R_{in}}$

Gain-Bandwidth Product (GBP):

- $GBP = A_v \times f_c$ (Gain $\times$ Cutoff Freq.)
- *Rule:* As single-stage gain increases, functional bandwidth decreases. Cascade stages to overcome this limitation.

STABILITY & NEGATIVE FEEDBACK

- **Frequency Domain Instability:** Identified by **Gain Peaking** (sharp spike in magnitude plot near cutoff frequency).
- **Time Domain Instability:** Identified by **Ringing** and **Overshoot** on a square wave step response.
- **Resolution:** Instability is caused by poles too close to the imaginary axis (low phase margin). Fix by adding a small **compensation capacitor** in parallel with the feedback resistor (adds a dominant pole).

DECOUPLING CAPACITORS

Real-world DC power lines contain noise. Capacitors are used to "clean" the signal.

- **Bulk Capacitors (e.g., $10\mu F$ electrolytic):** Filter low-frequency noise. Generally need 1 per board per power line. Can be placed anywhere on the rail.
- **Bypass Capacitors (e.g., $0.1\mu F$ ceramic):** Filter high-frequency noise. Must be placed *as close as possible* to the IC power pins. Need 1 per IC per power line.

LOADING EFFECT & IMPEDANCE

Ideal Component Impedances:

- Voltmeter / Oscilloscope: $Z_{in} = \infty$ (Open circuit)
- Ammeter: $Z_{in} = 0\Omega$ (Short circuit)
- Voltage Source: $Z_{out} = 0\Omega$

Loading Effect: Connecting a load forms a voltage divider: $V_{load} = V_{source}\left(\frac{Z_{in}}{Z_{out}+Z_{in}}\right)$.

- *Rule of Thumb:* To avoid voltage attenuation (signal loss), ensure $Z_{in} \gg Z_{out}$.

COMMON MODE REJECTION (CMRR)

Voltages & Gains:

- Differential Voltage (V_{DM}): $V_1 - V_2$
- Common Mode Voltage (V_{CM}): $\frac{V_1+V_2}{2}$
- Output Voltage: $V_{out} = A_{DM}V_{DM} + A_{CM}V_{CM}$

CMRR Calculation:

- Ratio: $CMRR = \frac{A_{DM}}{A_{CM}}$
- Decibels (dB): $CMRR_{dB} = 20\log_{10}\left(\frac{A_{DM}}{A_{CM}}\right)$
- *Concept:* High CMRR is desired to reject noise (like 60Hz wall interference) that appears simultaneously on both input lines.

EQUIPMENT & LAB PROCEDURES

Lab-Volt Power Supply:

- To provide $+V_{cc}$ and $-V_{cc}$, use a bridge connection: tie negative terminal of one supply to positive terminal of the other (creates COM).
- *Crucial:* Always set current limits to protect circuits from short-circuit damage!

Function Generator (50Ω Load):

- Assumes it is driving a 50Ω load.
- If circuit has high input impedance (e.g., op-amp), $V_{load} \approx V_{int}$. Output voltage will be roughly *double* the display unless using a 50Ω terminating resistor.

Tektronix Oscilloscope:

- **Probe Comp (x10):** Calibrate using test square wave.

Overcompensated: Leading edge spike.
Undercompensated: Rounded leading edge.
Properly Compensated: Flat, crisp square wave.

- **Amplitude Scaling:** Scale waveform to fill vertical display without clipping to maximize ADC resolution.

ANALOG-TO-DIGITAL (ADC)

- **Signal Conditioning:** Shift and scale a small, bi-polar input signal to fit an ADC's input range. Apply a **DC offset** to center the signal, then **amplification** to stretch it.
- **SNR (Ideal):** For an N-bit ADC, $SNR_{dB} \approx 6.02N + 1.76$.
- **Nyquist Theorem:** Sampling rate must be at least twice the maximum frequency ($f_s \geq 2f_{max}$) to avoid aliasing.

EXAMPLE PROBLEMS & SETUP

1. Complex Numbers & Polar Conversion

Prob: Given $Z_1 = 3 + j4$, $Z_2 = 5\angle 30°$, find $Z_1 \times Z_2$.
Sol:
1) Convert Z_1: $r = \sqrt{3^2 + 4^2} = 5$.
2) Angle: $\theta = \tan^{-1}(4/3) \approx 53.13° \implies Z_1 = 5\angle 53.13°$.
3) Multiply magnitudes, add angles: $5 \times 5 = 25$, $53.13° + 30° = 83.13°$.
Ans: $25\angle 83.13°$

2. s-Domain to Time-Domain

Prob: $V_{out}(s) = \frac{10}{s+500}$, find $v_{out}(t)$.
Sol:
1) Standard Pair: $\mathcal{L}^{-1}\left\{\frac{A}{s+a}\right\} = Ae^{-at}u(t)$.
2) Map variables: $A = 10$, $a = 500$.
Ans: $V_{out}(t) = 10e^{-500t}u(t)$ V

3. Complex Signals, Impedance, RMS

Prob: $v(t) = 10\sin(1000t)$ V across a $0.1\mu F$ cap. Find V_{RMS} and Z_C.
Sol:
1) RMS Voltage: $V_{RMS} = \frac{V_{peak}}{\sqrt{2}} = \frac{10}{\sqrt{2}} \approx$ **7.07 V_{RMS}**.

2) Identify frequency: $\omega = 1000$ rad/s.
3) Impedance: $Z_C = \frac{1}{j\omega C} = \frac{1}{j(1000)(0.1\times10^{-6})} = \frac{1}{j(10^{-4})}$.
Ans: $-$**j10 k**Ω (or $10\angle-90°$ kΩ).

4. Active Circuit Transfer Function
Prob: Derive $H(s) = \frac{V_{out}}{V_{in}}$ for ideal non-inverting op-amp (R_f feedback, R_{in} to ground).
Sol:
1) Ideal rules: $V_+ = V_-$ and $I_{in} = 0$.
2) Find node voltages: $V_+ = V_{in} \implies V_- = V_{in}$.
3) KCL at V_- node: $\frac{V_- - 0}{R_{in}} + \frac{V_- - V_{out}}{R_f} = 0$.
4) Sub V_{in}: $V_{in}\left(\frac{1}{R_{in}} + \frac{1}{R_f}\right) = \frac{V_{out}}{R_f}$.
Ans: Multiply by R_f: $\mathbf{H(s) = 1 + \frac{R_f}{R_{in}}}$.

5. Analog to Digital Converters
Prob: 10-bit ADC, $0V$ to $5V$ range. Find step size.
Sol:
1) Total discrete states: $2^n = 2^{10} = 1024$.
2) Voltage Range: $V_{max} - V_{min} = 5V - 0V = 5V$.
3) Step size: $\frac{\text{Range}}{\text{States}} = \frac{5V}{1024}$.
Ans: $\approx$ **4.88 mV**.

6. Common Mode Rejection Ratio (CMRR)
Prob: An instrumentation amp has a differential gain $A_{DM} = 100$ and a common mode gain $A_{CM} = 0.01$. Find the CMRR in dB.
Sol:
1) CMRR Ratio: $\frac{A_{DM}}{A_{CM}} = \frac{100}{0.01} = 10000$.
2) Convert to dB: $20\log_{10}(10000) = 20(4)$.
Ans: 80 dB.

7. Coupled Op-Amps Transfer Function
Prob: Derive $H(s)$ for two cascaded op-amps (e.g., Stage 1: Inverting, Stage 2: Non-Inverting).
Foundational Setup:
1) Identify individual transfer functions: $H_1(s) = -\frac{R_{f1}}{R_{in1}}$ and $H_2(s) = 1 + \frac{R_{f2}}{R_{in2}}$.
2) Apply the cascade rule: $H_{total}(s) = H_1(s) \times H_2(s)$.
3) *Self-Check:* What happens to the overall phase if both stages are inverting?

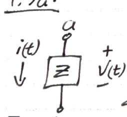
8. Prelab 4, Problem 1 (LPF)
Prob: Find an expression for cutoff frequency f_c. Given $C_3 = 10\mu F$, find f_c for $R_3 = [100\Omega, 1k\Omega, 10k\Omega, 100k\Omega]$.
Foundational Setup:
1) Foundation: First-order RC filter cutoff is $f_c = \frac{1}{2\pi RC}$.
2) *Self-Check:* Plug in the provided values. Remember μF is 10^{-6}!

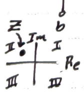
9. Lecture 4a: ADC & Stability
Prob: Determine quantization step size and resolve ringing.
Foundational Setup:
1) Step size (ΔV) = $\frac{V_{range}}{2^N}$.
2) *Stability Check:* Ringing/peaking means poles are too close to the imaginary axis. Resolve by adding a parallel compensation capacitor to the feedback resistor.

10. HW 1: Problem 3b (Phasor Ops)
Prob: Given $f(t) = 12\cos(200t + 45°)$ and $g(t) = 5\sin(200t + 102°)$, find $h_b(t) = f(t)\cdot g(t)$.
Foundational Setup:
1) Convert sine to cosine: $\sin(\theta) = \cos(\theta - 90°) \implies g(t) = 5\cos(200t + 12°)$.
2) Convert to phasors: $F = 12\angle45°$, $G = 5\angle12°$.
3) Multiply magnitudes, add phases, and convert back to time-domain.

11. HW 1: Problem 5 (Power Calculations)
Prob: Given $v(t) = 12\cos(\omega t + 20°)$ and $Z = -1.6971 + j1.6971\ \Omega$. Find current $i(t)$, real power P, reactive power Q, and apparent power $|S|$.
Foundational Setup:
1) Convert Z to polar form: $r = \sqrt{x^2+y^2}$, $\theta = \tan^{-1}(y/x)$. *Hint: Watch your quadrant!*
2) Phasor Current: $I = V/Z$. Remember to divide magnitudes and subtract angles. Convert the resulting phasor back to time-domain $i(t)$.
3) Apparent Power: $|S| = V_{RMS}I_{RMS} = \frac{1}{2}V_{peak}I_{peak}$.
4) Real / Reactive Power: $P = |S|\cos(\theta_z)$, $Q = |S|\sin(\theta_z)$, where θ_z is the angle of the impedance.

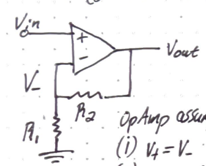
1.3b:
$f(t) = 12\cos(200t + 45°)$
$g(t) = 5\sin(200t + 102°)$
c) $h_b(t) = f(t)\cdot g(t)$
$\vec{F} = 12\angle45° = 12e^{j45°}$
$= 12[\cos(45°) + j\sin(45°)]$
$= 12\cos(45°) + j\,12\sin(45°)$
$= 8.4853 + j\,8.4853$
$g(t) = 5\cos(200t + 102° - 90°)$
$= 5\cos(200t + 12°)$
$= 5\angle12°$
$= 5\cos(12°) + j\,5\sin(12°)$
$= 4.8907 + j\,1.0396$
$f(t)\cdot g(t) \rightarrow \vec{F} * \vec{G} = (12\cdot5)\angle(45° + 12°)$
$= 60\angle57°$
$h_b(t) = 60\cos(200t + 57°)$

1.5a:


$V(t) = 12\cos(\omega t + 20°)$
$Z = -1.6971 + j1.6971$
$\vec{V} = 12\angle20°$
$Z = \sqrt{Re^2 + Im^2}\angle\tan^{-1}\left(\frac{Im}{Re}\right)$
$= \sqrt{-1.6970^2 + 1.6971^2}\angle\tan^{-1}\left(\frac{Im}{Re}\right)[\pm180°]$
$= 2.4\angle135°$


$\vec{I} = \frac{\vec{V}}{Z} = \left(\frac{12}{2.4}\right)\angle(20° - 135°) = 5\angle-115°$
$i(t) = 5\cos(\omega t - 115°)$

LS 4a:
ADC # of bits: N
# of levels = 2^N
Input voltage range
Q: How many volts does each level's step-size represent?
$N = 8$ & input voltage range = $[-1V, +1V]$
$\frac{\text{Volts}}{\text{level}} = \frac{\text{voltage range}}{\text{\# of levels}} = \frac{2V}{256\text{ levels}} \approx 7.8\ \text{mV/level}$
$2^8 = 256$ $\quad$ $2V$
Q: What voltage is represented by
0000 0110 ?
$= 6$
6 levels $* 7.8$ mV/level $= 46.8$ mV (from the bottom)
$-1v + 0.0486V = \boxed{-0.9532V}$

PL 4.1 Derrive the transfer fn.
$H(j\omega) = V_{out}/V_{in}$


$V_{out} = V_{in}^*$
$V_- = V_{out}^*$
$V_{out}^* = = \frac{R_L}{R_S + R_L}V_S$
$= \frac{R_1}{R_2 + R_1}V_{in}^*$
OpAmp assumptions:
(i) $V_+ = V_-$
(ii) $i_- = 0 = i_+$
$V_{in} = \frac{R_1}{R_1 + R_2}V_{out}$
$\frac{R_1 + R_2}{R_1} = V_{out}/V_{in}$
$= \frac{R_1}{R_1} + \frac{R_2}{R_1} = \boxed{1 + \frac{R_2}{R_1} = H(j\omega)}$

2.5, 2.6
$R = 560\ \Omega$ $\quad$ $i(0) = 0A$
$L = 100$ mH $\quad$ $i'(0) = 1000$ A/s
$C = 0.1\mu F$
$I(s) = \frac{i(0)s + [i'(0) + \frac{R}{L}i(0)]}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$
$a = 5600$ $\quad$ $b = 100\cdot10^6$
$s_{1,2} = \frac{-a \pm \sqrt{a^2 - 4b}}{2} = -\left(\frac{R}{2L}\right) \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$
$= -2800 \pm j9600$
$s^2 + as + b = (s + c)^2 + d$
$c = \frac{a}{2} = \frac{R}{2L}$ & $d = b - \frac{a^2}{4} = \left[\frac{1}{LC} - \left(\frac{R}{2L}\right)^2\right]$
$\frac{i(0)s + [i'(0) + \frac{R}{L}i(0)]}{"(s + R/2L)^2 + [1/LC - (R/2L)^2]"} = A\left[\frac{s + R/2L}{"}\right] + B\left[\frac{\sqrt{\frac{1}{LC} - (R/2L)^2}}{"}\right]$
$i(0)s + [i(0) + \frac{R}{L}i(0)]s^0 = A_s + \left[A\left(\frac{R}{2L}\right) + B\sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}\right]s^0$
$i(0) = A = 0$
$i'(0) + \frac{R}{L}i(0) = A\left(\frac{R}{2L}\right) + B\sqrt{\frac{1}{LC} - \left(\frac{R}{2}\right)^2}$
$i'(0) = 1000 = B\,9600$
$B = \frac{1000}{9600} \approx 0.1042$
$I(s) = 0.1042\left[\frac{9600}{(s + 2800)^2 + 9600^2}\right]$
Inverse Laplace Fn (#19 on slide)
$\boxed{i(t) = 0.1042e^{-2800t}\sin(9600t)\,A \quad t \geq 0}$

TAC